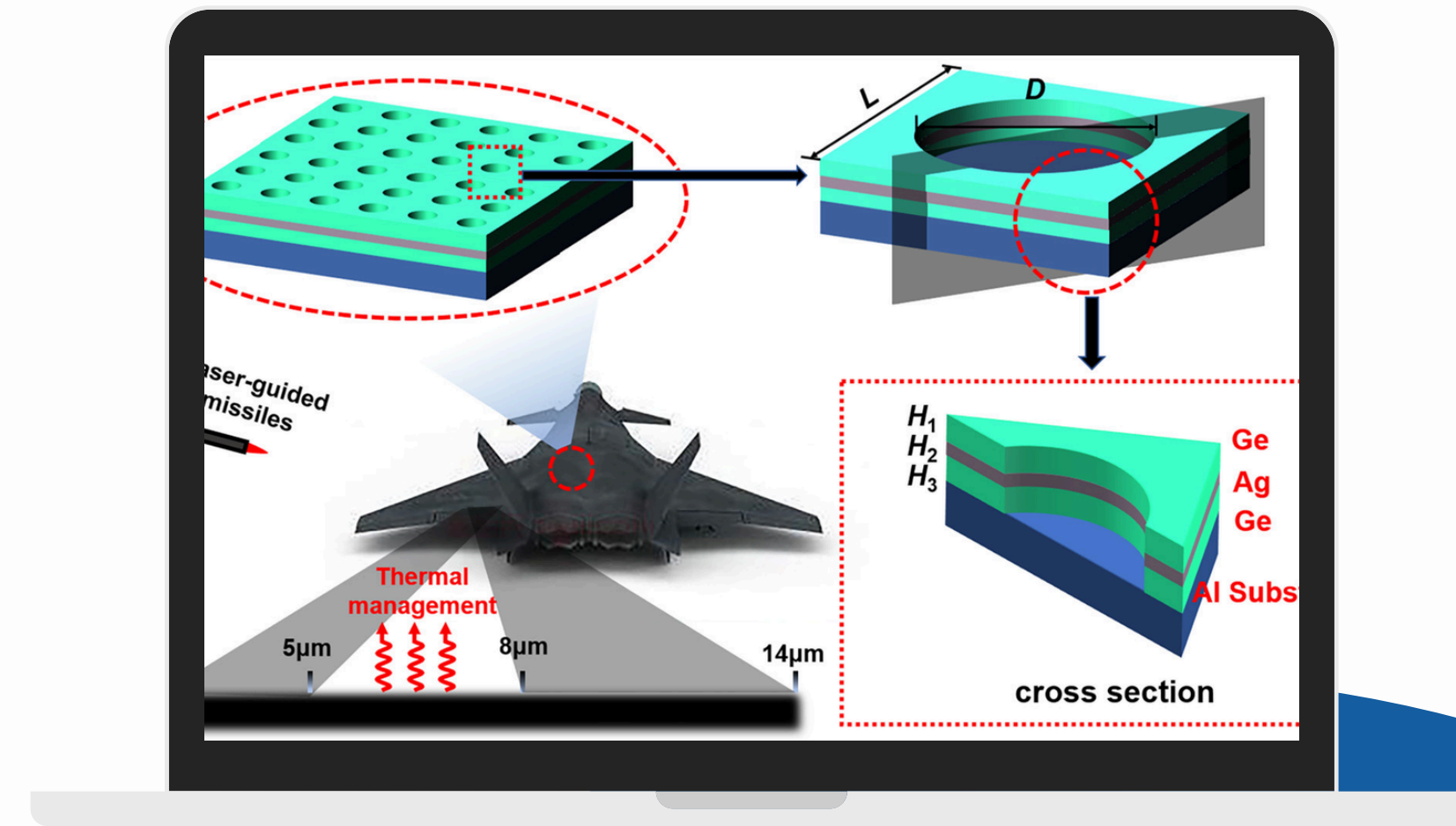




Design and Simulation of Metasurface-based absorber for 5th generation Stealth Aircraft

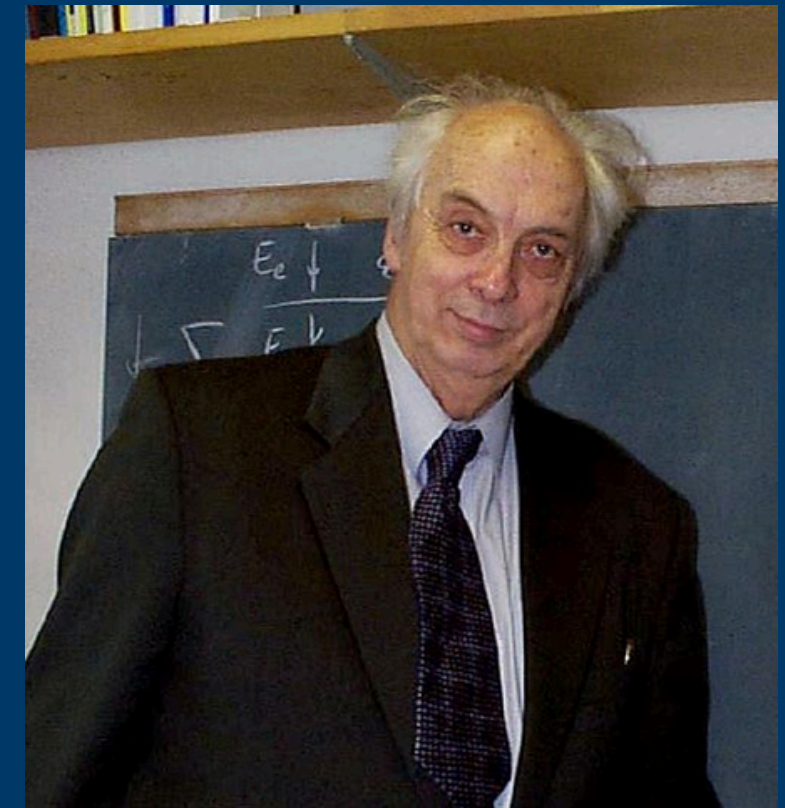
Project Group : 1	
Name	Roll no.
Ayan Munshi	35000323063
Anamika Sharma	35000323061
Aditya Kumar	35000323060
Hritik Giri	35000323066
Priya Mahanty	35000322029
Choya Mondal	35000322070

Under supervision of Dr. Hiranmay Mistri



CONTENT

▶▶	INTRODUCTION OF STEALTH TECHNOLOGY	01
▶▶	LITRATURE MODEL	02
▶▶	OUR MODEL	03
▶▶	CONCLUSION	04
▶▶	REFERENCE	05
		06



VG VESELAGO

First Study Of The effect of a negative permittivity and permeability on the wave propagation

Introduction Of Stealth technology

The cornerstone of radar-based stealth is the minimization of an object's Radar Cross Section (RCS), which is a measure of how detectable an object is by a radar system.

Problem Statement:-

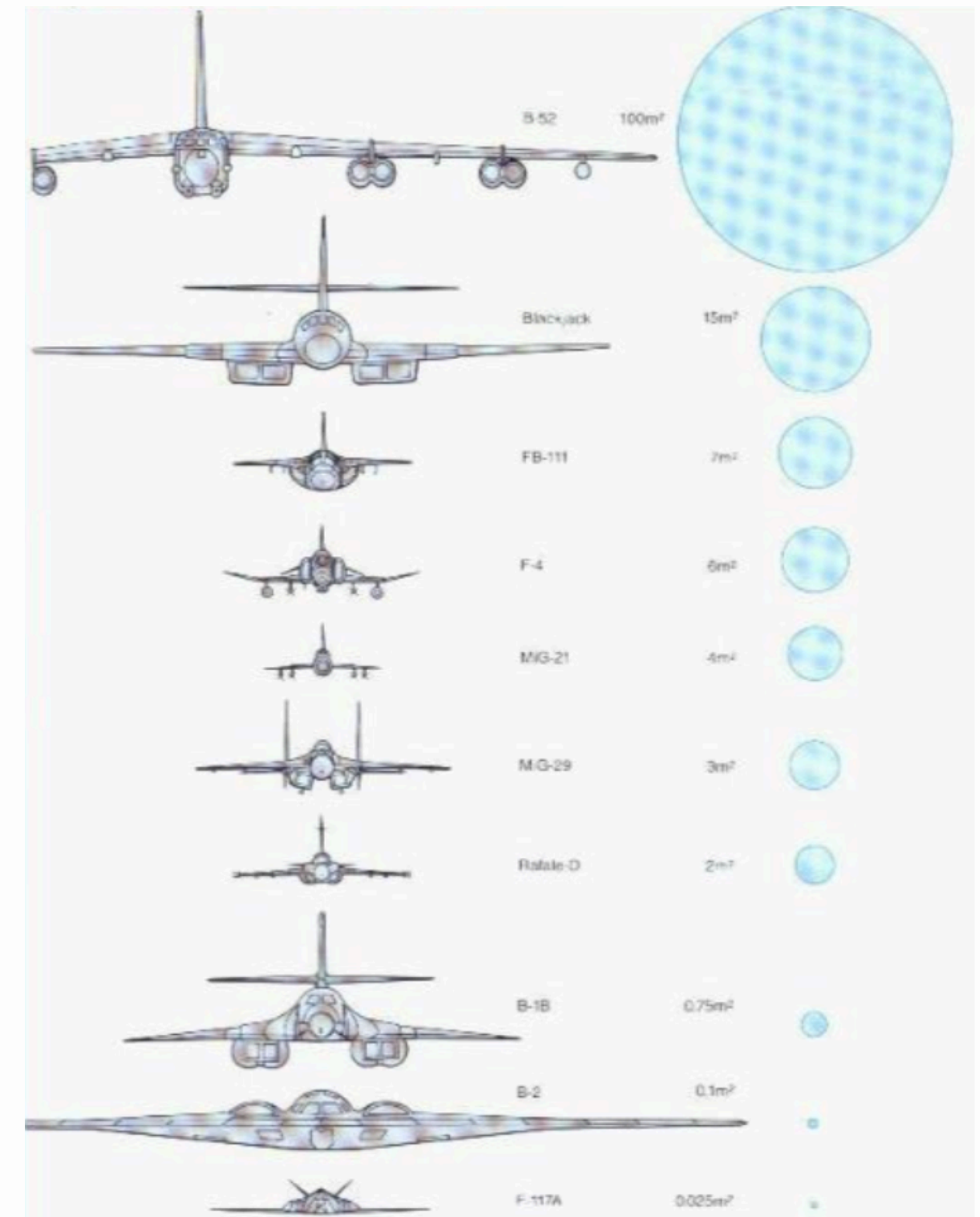
- **The Modern Threat: High-Frequency Radar** Modern detection and imaging systems are moving into the THz frequency band
- **Traditional Stealth is too Bulky**
- **Material Roadblocks**

Objective of This Work:-

- **lightweight Multi-Band Graphene-Based Metamaterial Absorber** operating in the THz regime.
- **Near-unity (>90%) multi-band absorption.**
- **Excellent bistatic and monostatic Radar Cross Section (RCS) reduction** across the target THz bands.

The Solution:-

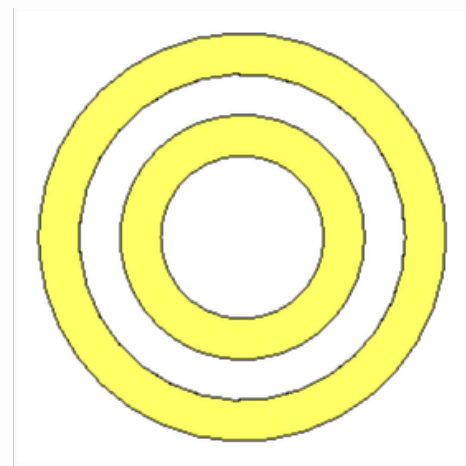
- **Ultra-Thin Profile**
- **Multi-Band Performance**
- **Smart Radar Evade**



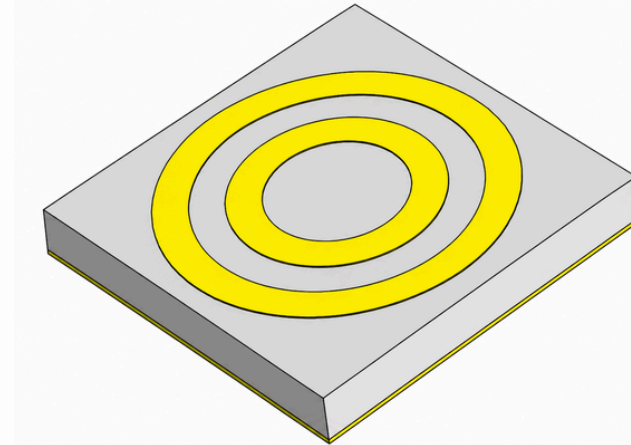
Literature Survey:-

We design and simulate the graphene based metamaterial absorber (MMA) based on dual-ring structure at terahertz frequencies using CST Studio software. This structure demonstrates perfect absorption at 1.6 and 2.89 THz

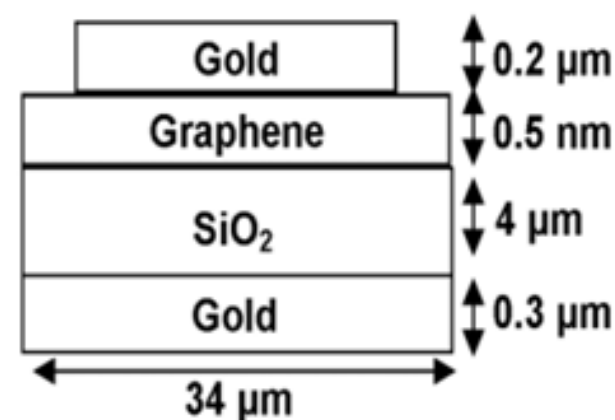
- Structure And Modeling Method



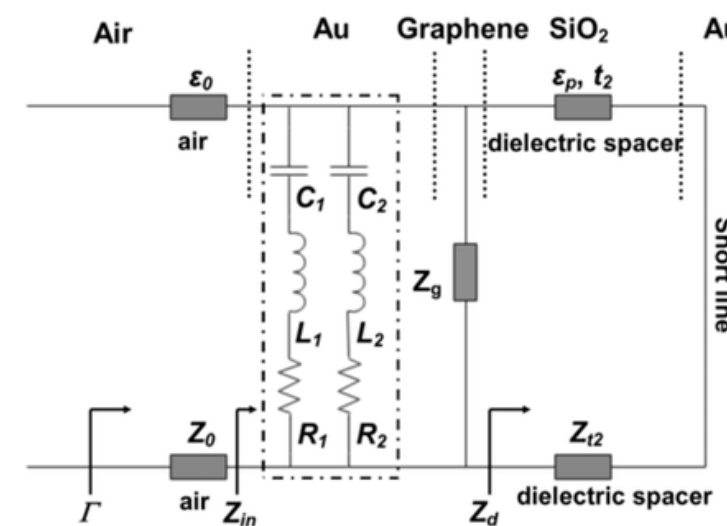
[A]



[B]



[C]



[D]

- **Ground Plane (bottom Layer)**

Material:- Gold(lossymetal)
Conductivity:- 4.09×10^7
Thickness:- $0.3 \mu\text{m}$

- **Substrate Plane (middle Layer)**

Material:- SiO₂(dielectric)
Loss Tangent:- 0.001
Thickness:- $4 \mu\text{m}$
Epsilon:- 2.35

- **Graphene Plane (Third Layer)**

Material:- Graphene
Relaxation Time:- 0.1ps
Chemical Potential- 0.75 eV
Thickness:- 1nm

- **Gold Plane (Top Layer)**

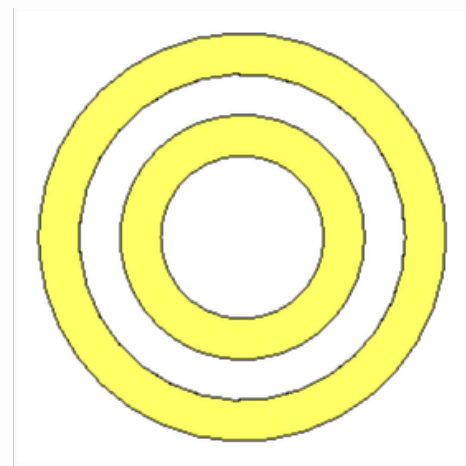
Thickness:- $0.2 \mu\text{m}$

Fig. [A][B][C] P. Jain, S. Garg, A. K. Singh, S. Bansal, K. Prakash, N. Gupta, A. K. Singh, N. Sharma, S. Kumar, and N. Sardana, "Dual Band Graphene Based Metamaterial Absorber for Terahertz Applications," in 2018 IEEE 13th Nanotechnology Materials and Devices Conference (NMDC), 2018

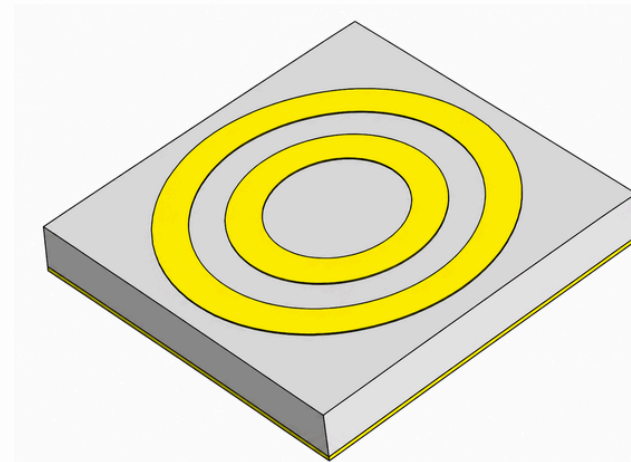
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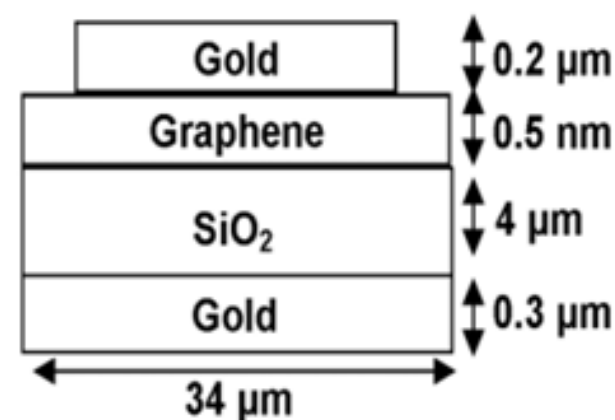
- Structure And Modeling Method



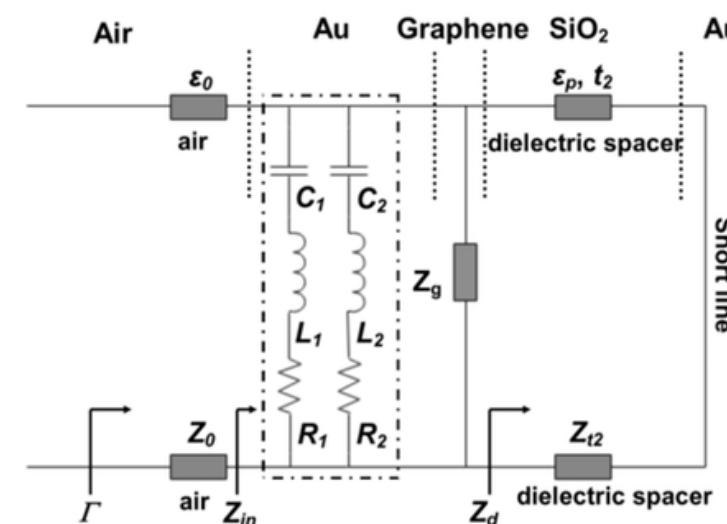
[A]



[B]



[C]



[D]

- **Ground Plane (bottom Layer)**

Material:- Gold(lossymetal)
Conductivity:- 4.09×10^7
Thickness:- $0.3 \mu\text{m}$

- **Substrate Plane (middle Layer)**

Material:- SiO2(dielectric)
Loss Tangent:- 0.001
Thickness:- $4 \mu\text{m}$
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- **Graphene Plane (Third Layer)**

Material:- Graphene
Relaxation Time:- 0.1ps
Chemical Potential- 0.75 eV
Thickness:- 1nm

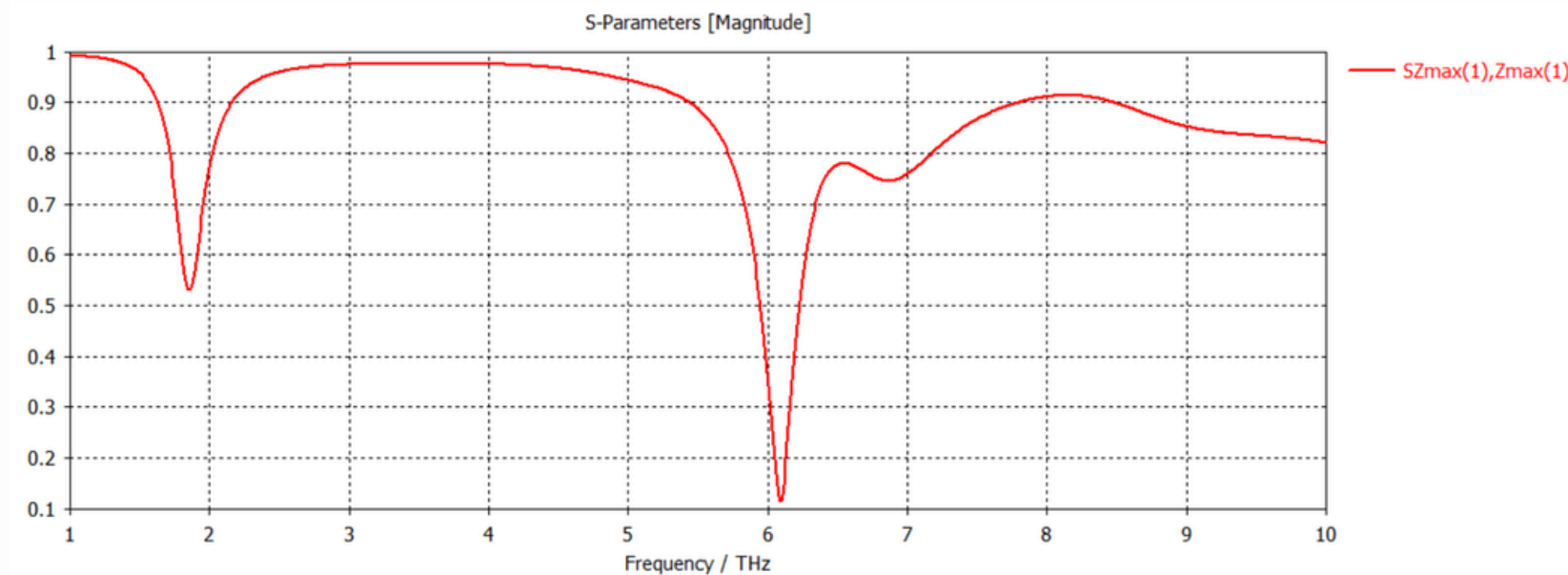
- **Gold Plane (Top Layer)**

Thickness:- $0.2 \mu\text{m}$

Fig. [A][B][C] P. Jain, S. Garg, A. K. Singh, S. Bansal, K. Prakash, N. Gupta, A. K. Singh, N. Sharma, S. Kumar, and N. Sardana, "Dual Band Graphene Based Metamaterial Absorber for Terahertz Applications," in 2018 IEEE 13th Nanotechnology Materials and Devices Conference (NMDC), 2018

Literature Model Review

- Model Analysis



[1]

Fig.[1] S11 Parameter:-

Absorption Breakdown For The Red Curve ($A = 1 - |S_{11}|^2$)

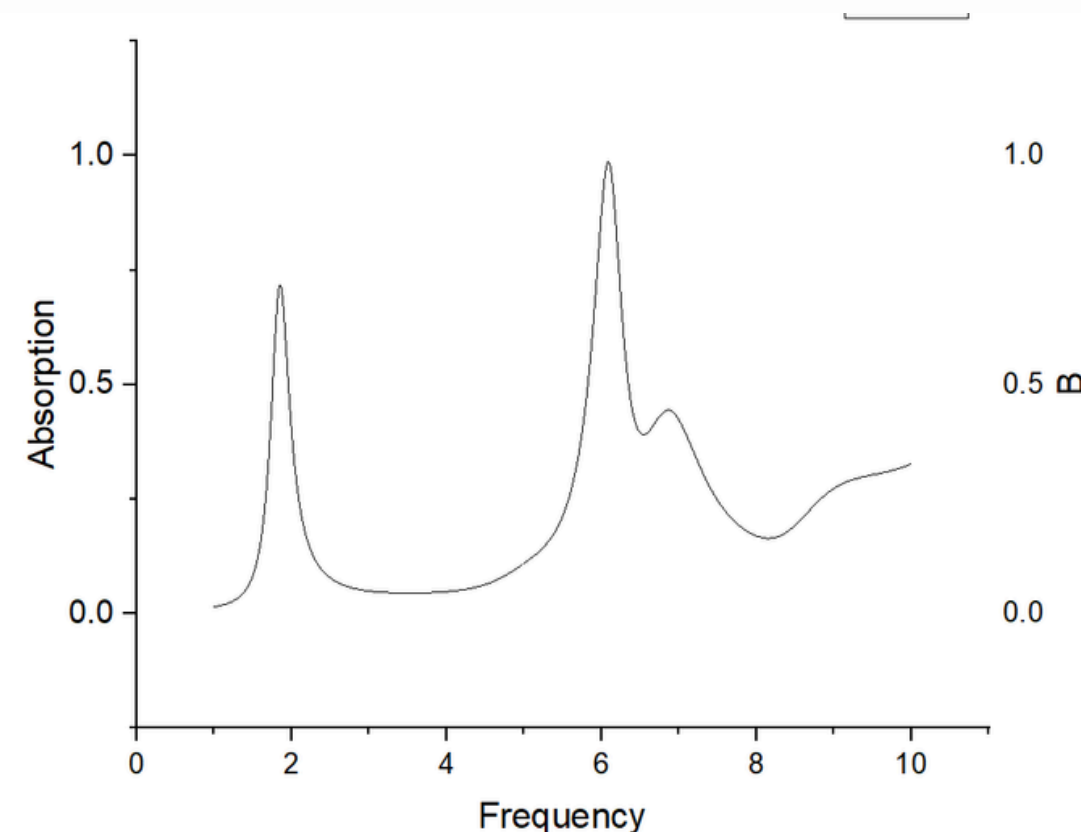
- Peak 1(at ~ 1.8THz): S11 Drop to 0.54 ==> 70% Absorption
- Peak 2(at ~ 6.1THz): S11 Drop to 0.14 ==> 98% Absorption

Physical Analysis of This Run:-

- Dual Band THz Operation
- Excellect Impedance Matching at 6THz
- Moderate Low Frequency Resonance

Fig.[2] Absortion vs Frequency Graph:-

- The Low-Frequency Band (1.9 THz)
- The High-Performance Band (6.3 THz with 7.0 THz Shoulder)

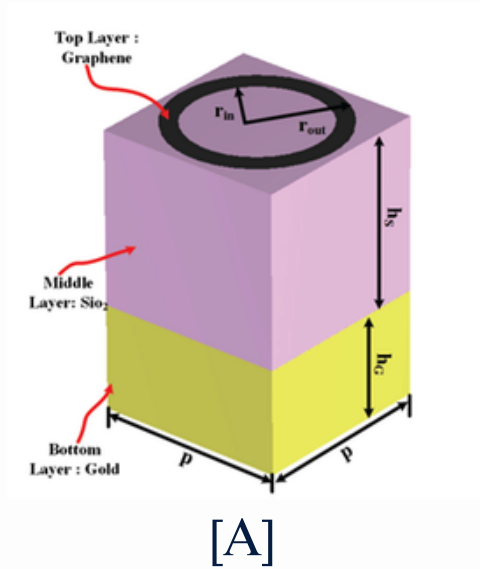


[2]

Literature Survey:-

This paper presents a broadband terahertz (THz) metamaterial absorber. The unit cell structure of this absorber consists of three layers. At the center frequency of 2THz, this circular graphene ring has a fractional bandwidth of 67% and absorptivity of more than 90%

- Structure And Modeling Method



Symbols	Values (μm)	Symbols	Values (μm)
h_s	26	r_{in}	7
h_G	1	r_{out}	9
p	20	t_G	0.1

- **Ground Plane (bottom Layer)**

Material:- Gold(lossymetal)
Conductivity:- 4.09e7
Thickness:- 1um

- **Substrate Plane (middle Layer)**

Material:- SiO2(dielectric)
Loss Tangent:- 0.001
Thickness:- 26um
Epsilon:- 2.25

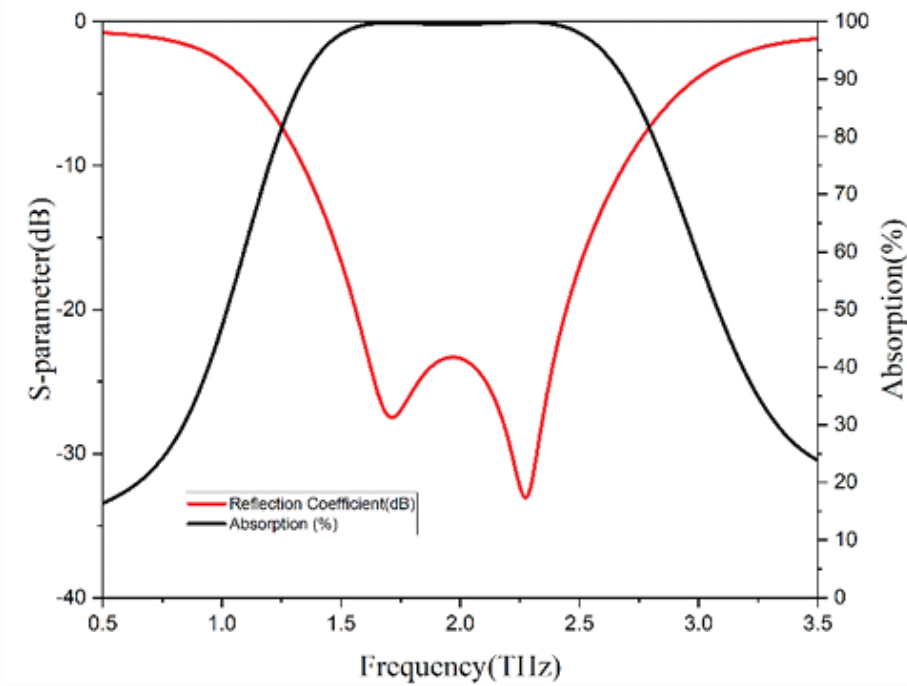
- **Graphene Plane (Third Layer)**

Material:- Graphene
Relaxation Time:- 0.1ps
Chamical Potential- 0.75 eV
Thickness:- 1nm

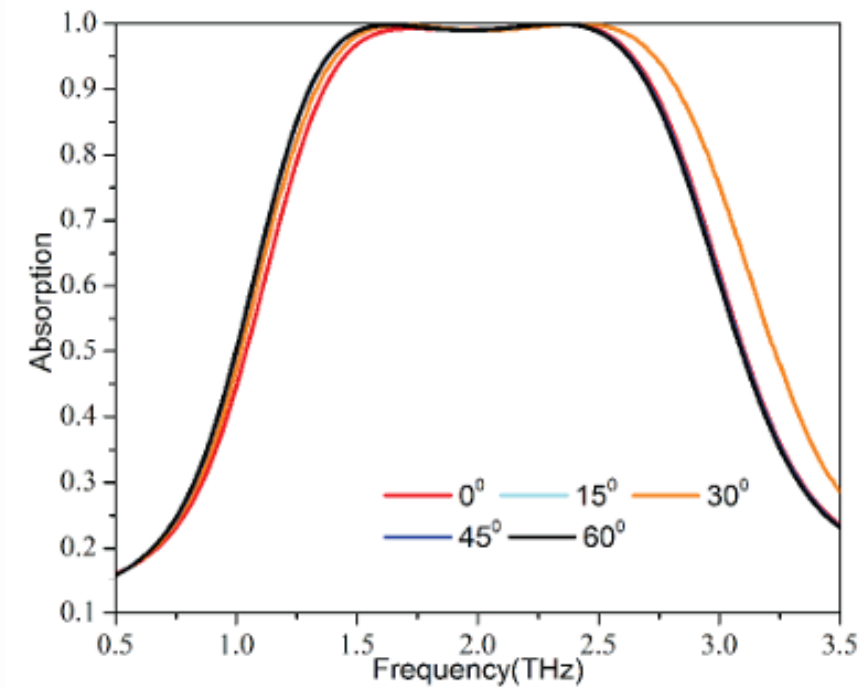
Fig. [A] L. N. Deekonda, S. K. Sahu, and A. K. Panda, "A Graphene-Based Broadband Metamaterial Absorber," in 2021 Advanced Communication Technologies and Signal Processing (ACTS), 2021, pp. 1-4, doi: 10.1109/ACTS53447.2021.9708250.

Literature Model Review

- Model Analysis



[1]



[2]

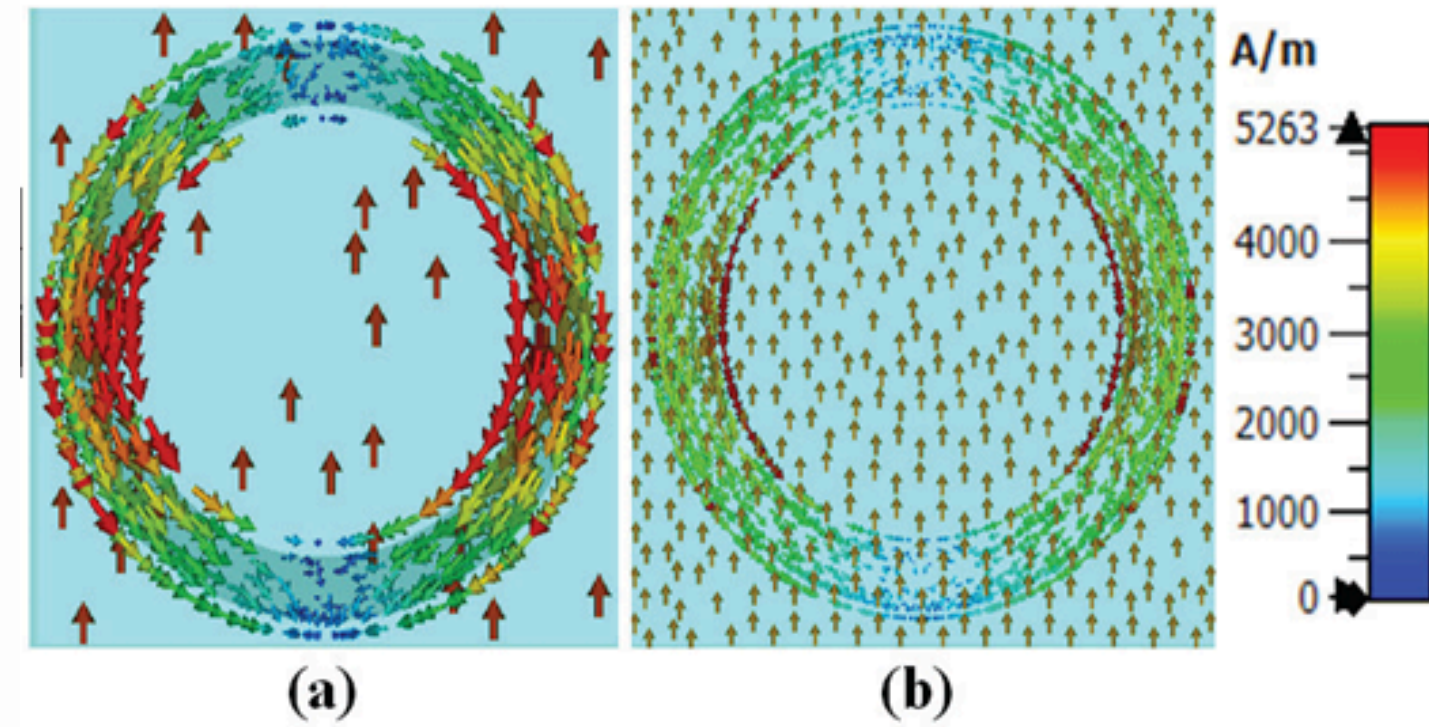


Fig. 3 Surface current of metamaterial at 1.8THz (a) top view (b) bottom view.

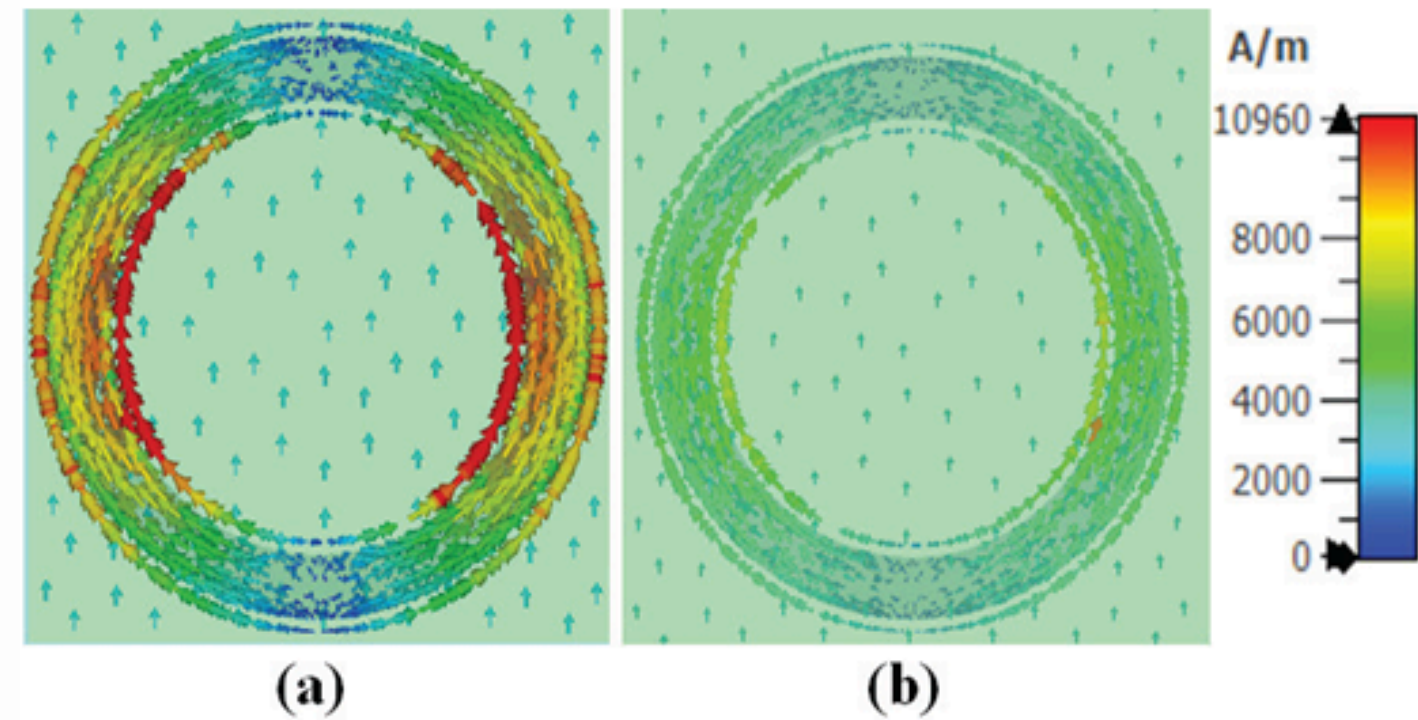
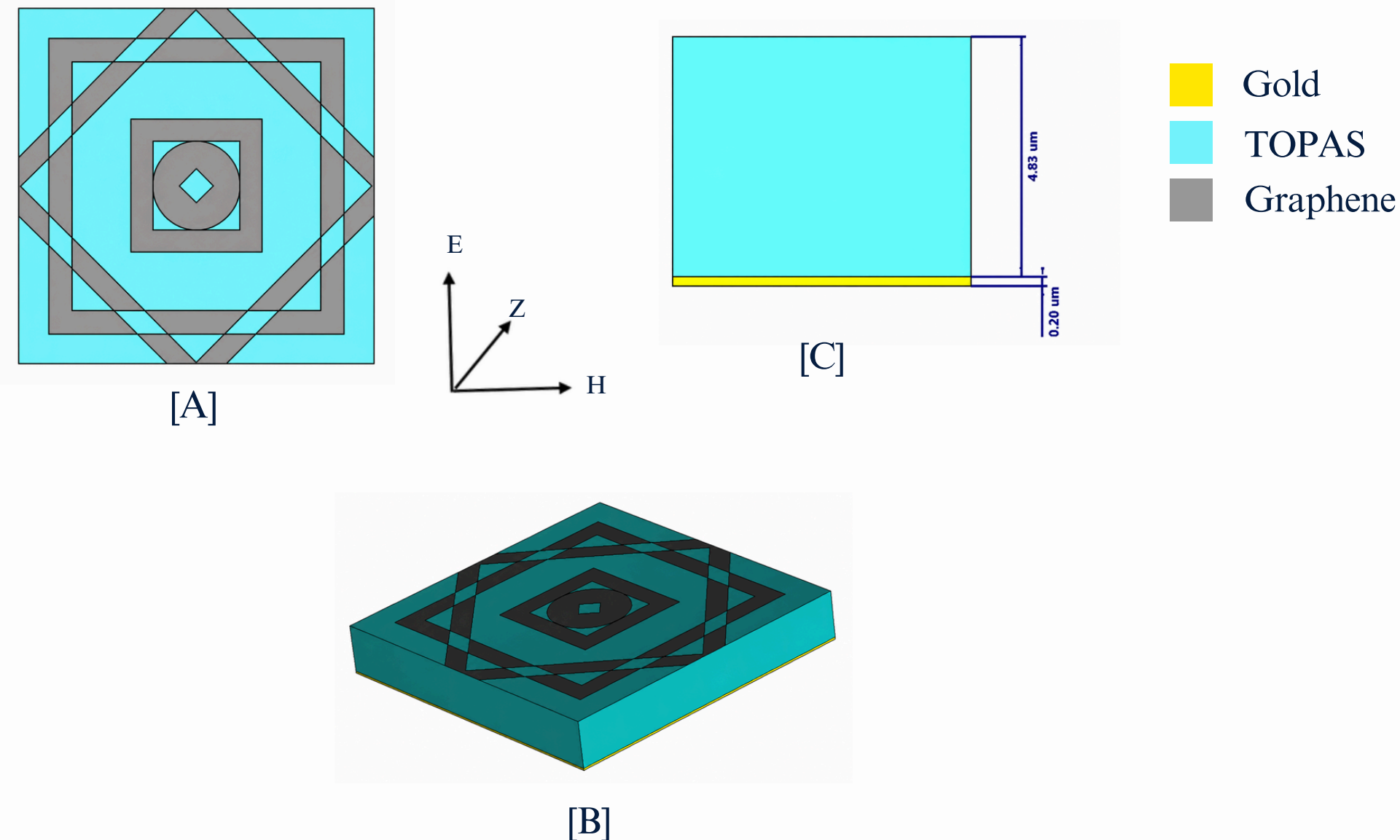


Fig. 4 Surface current of metamaterial at 2.7THz (a) top view (b) bottom view.

Model Review

We Designe a Graphene Based Metamaterial Unit Cell With Unique Resonator Structure For THz Frequency Absorber. Which is Absorbe Multi-band Terahertz Frequencys 8 THz to 20 THz.

- Structure And Modeling Method



- **Ground Plane (bottom Layer)**

Material:- Gold(lossy)
Conductivity:- $4.09e7$
Thickness:- 0.2 μm

- **Substrate Plane (middle Layer)**

Material:- TOPAS(dielectric)
Loss Tangent:- 0.001
Thickness:- 4.83333 μm
Epsilon:- 2.35

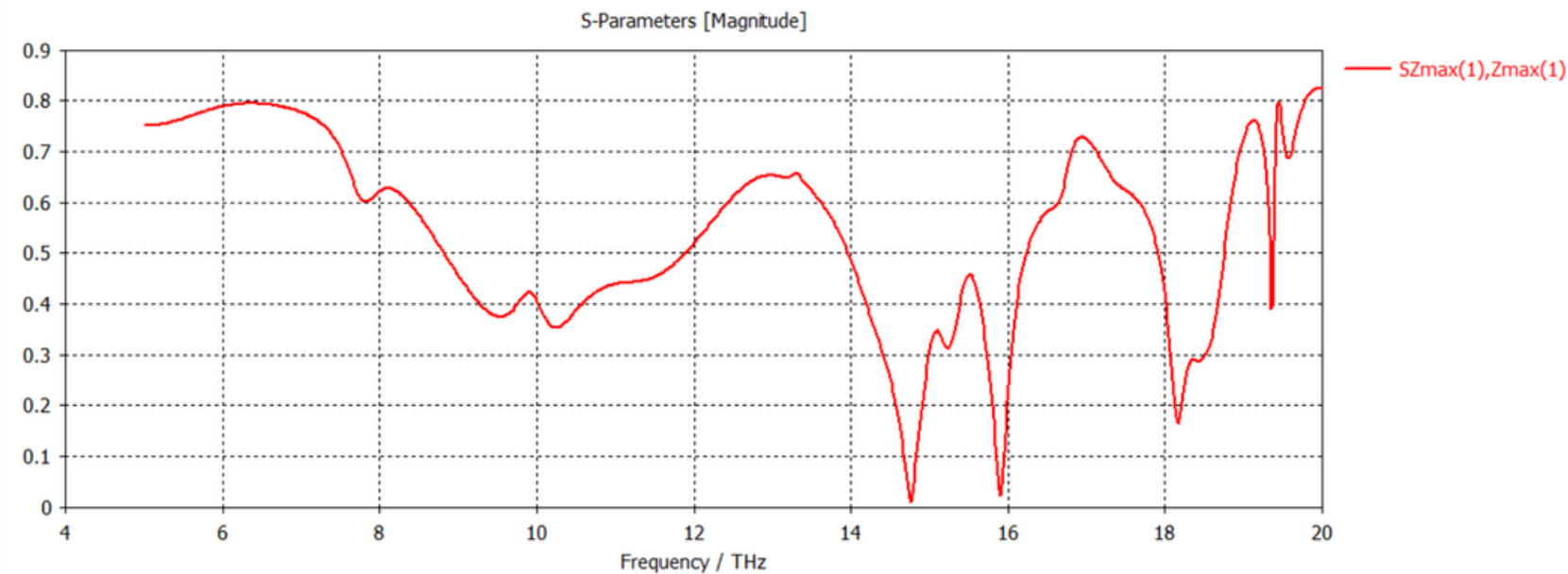
- **Graphene Plane (Top Layer)**

Material:- Graphene
Relaxation Time:- 0.1ps
Chemical Potential- 0.75 eV
Thickness:- 1nm

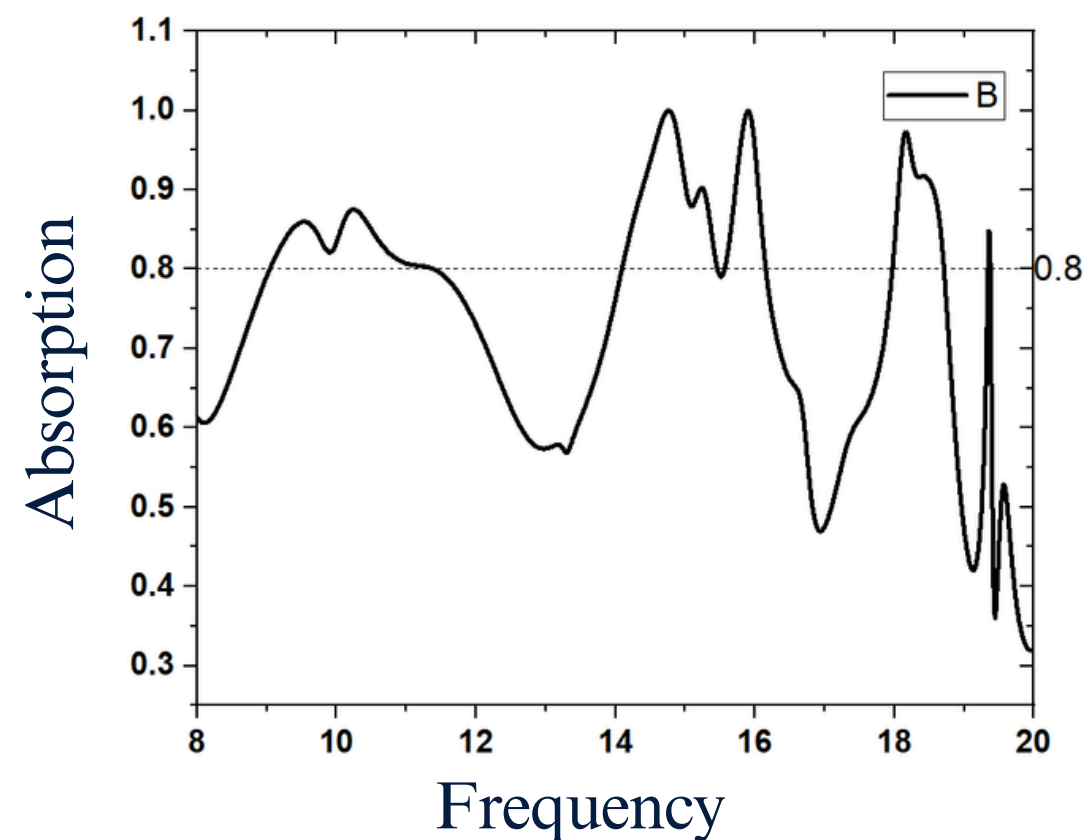
Fig. [A][B][C] 3D view, top view and Side View of Structure diagram

Model Review

- Model Analysis



[1]



[2]

Fig.[1] S11 Parameter:-

Absorption Breakdown For The Red Curve ($A = 1 - |S_{11}|^2$)

- Peak 1(at ~ 14.8THz): S11 Drop to 0.01 ==> 98% Absorption
- Peak 2(at ~ 16.0THz): S11 Drop to 0.01 ==> 96% Absorption
- Peak 3(at ~ 19.6THz): S11 Drop to 0.35 ==> 87.8% Absorption

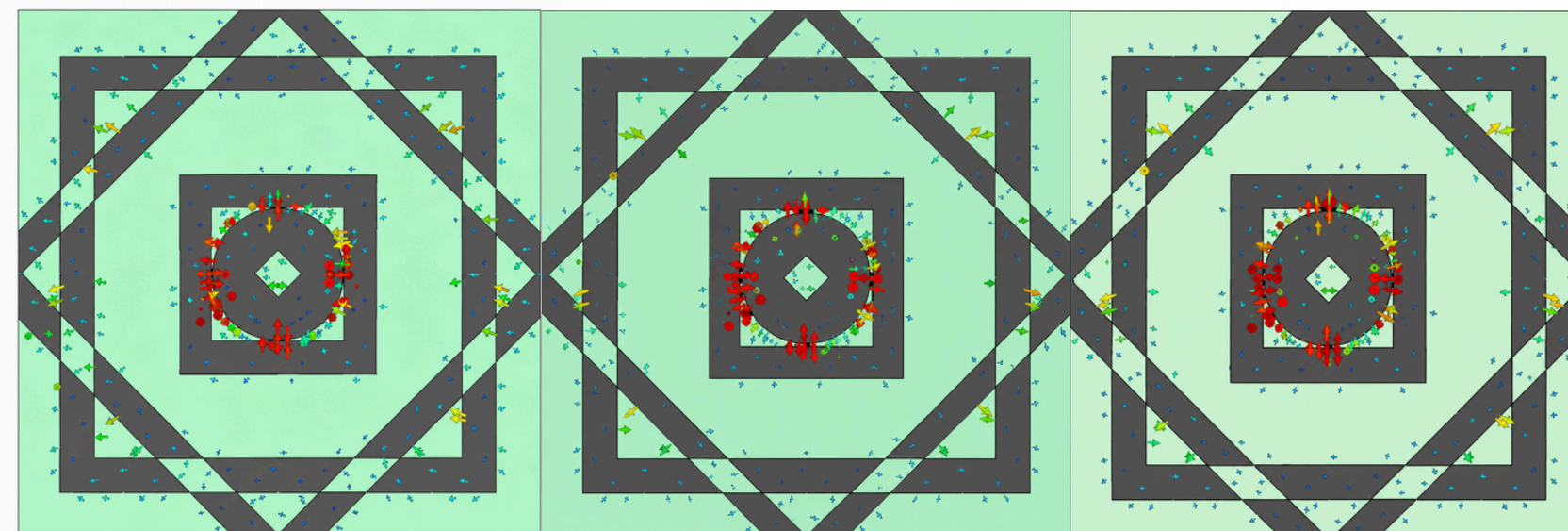
We have successfully simulate a multi-band Terahertz Absorber with almost flawless impedance matching.

Fig.[2] Absorption vs Frequency Graph:-

- Band 1: The Broadband Absorption Window (9.0 THz- 11THz)
Key peaks:
 - Peak 1(at ~ 9.5 THz): 86% absorption
 - Peak 2(at ~ 10.2 THz): 87% absorption
- Band 2: The Dual-Peak “Near Unity” Window (14.0 THz- 16.3THz)
Key peaks:
 - Peak 3(at ~ 14.8 THz): 99.8% absorption
 - Peak 4(at ~ 16.0 THz): 99.6% absorption
- Band 3: The High Frequency Peak Window (18.1 THz- 18.9THz)
Key peaks:
 - Peak 5(at ~ 18.3 THz): 96.0% absorption

Model Review

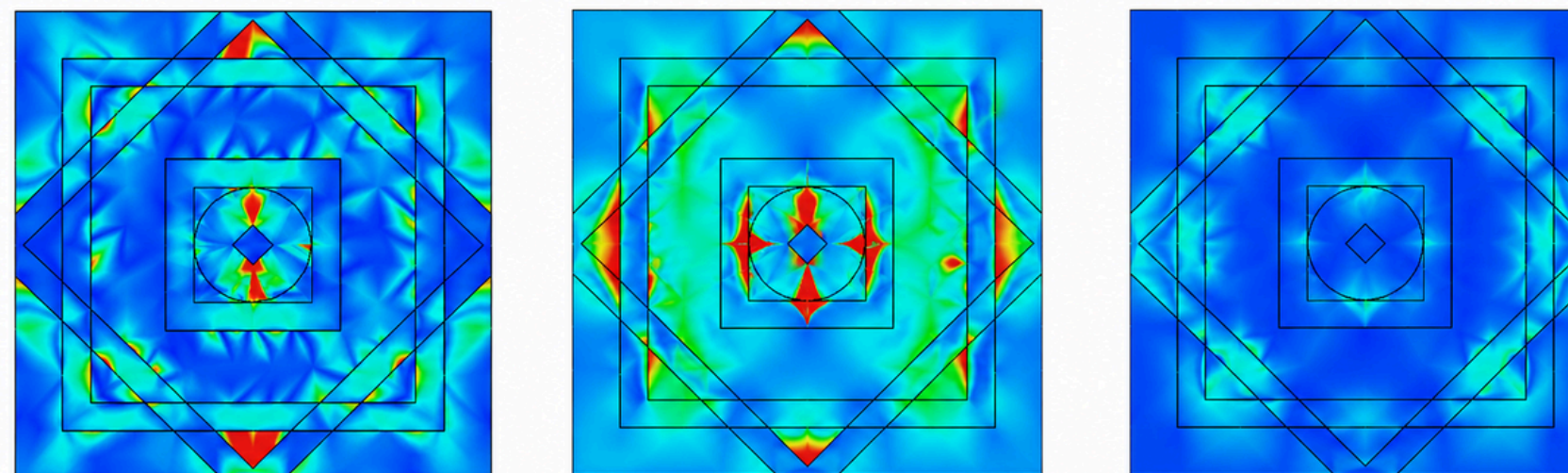
- Model Surface current & Heatmaps



[3]

[4]

[5]



[6]

[7]

[8]

Fig.[1][2][3] Visual Breakdown Of the Simulation:-

in this Geometry the vector arrows(Surface Current)

- **ColorScal** :Red/Orange indicates high current density, while green and blue indicate low current density (weak and non-resonant)
- inner circular ring is Highly resonant

Surface Current:-

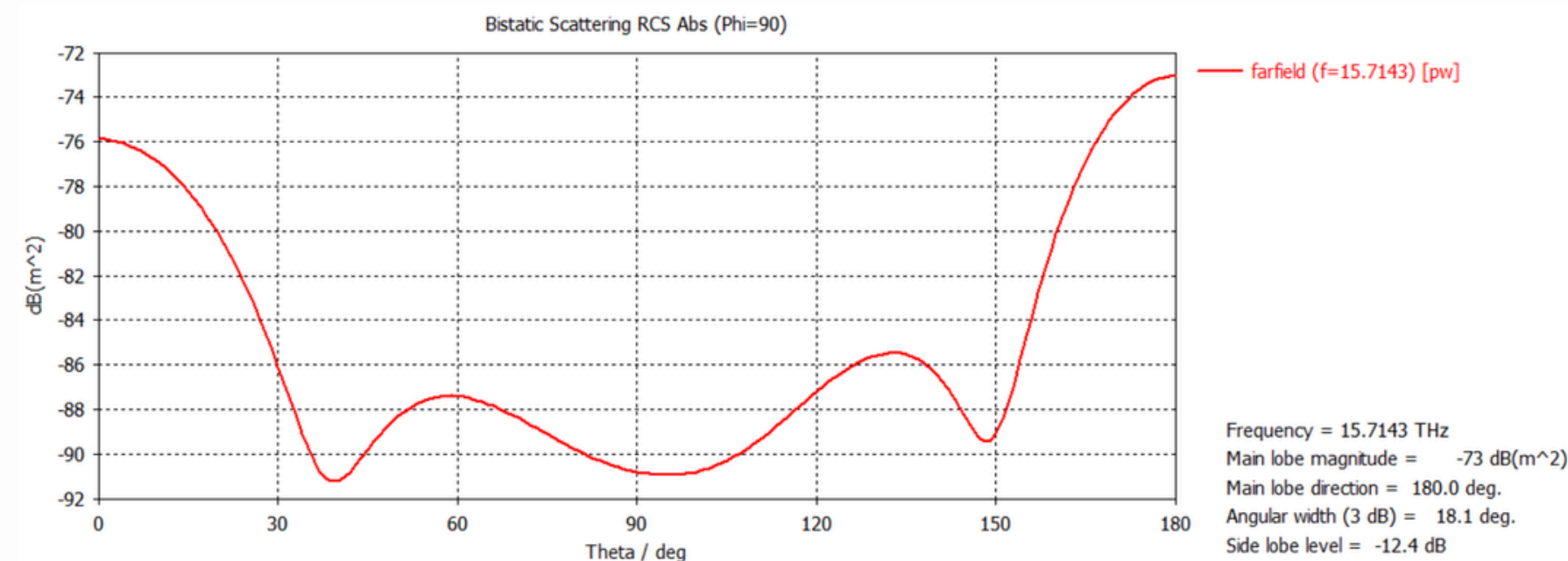
- **Resonance & Absorption**
- **Energy Dissipation**

Fig.[6][7][8] Magnitude contour plots(Heatmaps):-

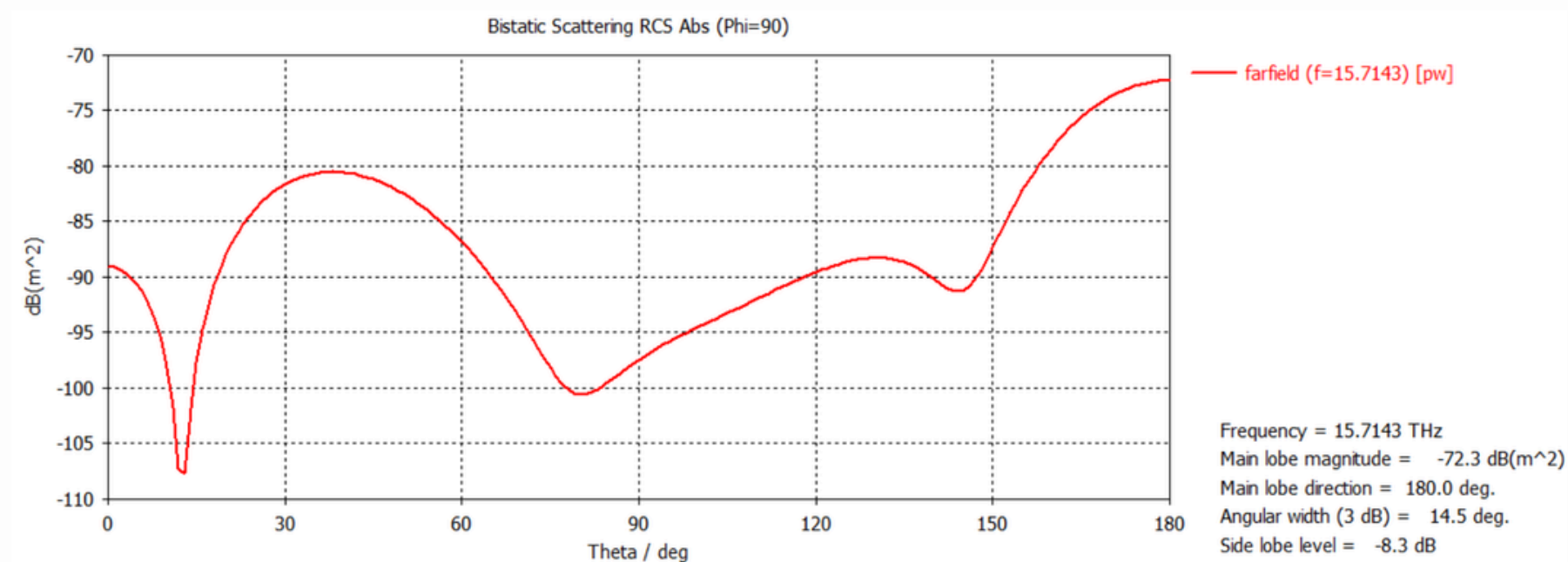
While your previous vector plots used arrows to show the direction of current flow, these heatmaps show exactly where the electromagnetic energy is concentrated. Red/orange indicates peak energy concentration (hotspots), while blue indicates areas with little to no electromagnetic activity.

Model Review

- **Bi Static RCS(Radar Cress Section):-** To evaluate a metamaterial's performance, simulate a PEC plate of the identical physical dimensions as a baseline. The difference in their scattered fields, calculated as $RCS(pec)-RCS(meta)$, directly quantifies the metamaterial's RCS reduction.



[9]



[10]

Fig.[9] This is shows the RCS of PEC Unit cell

- A PEC is an idealized material that reflects 100% of incident electromagnetic waves with absolutely zero energy absorption.
- In stealth technology research, this graph serves as the mandatory experimental baseline (control group).

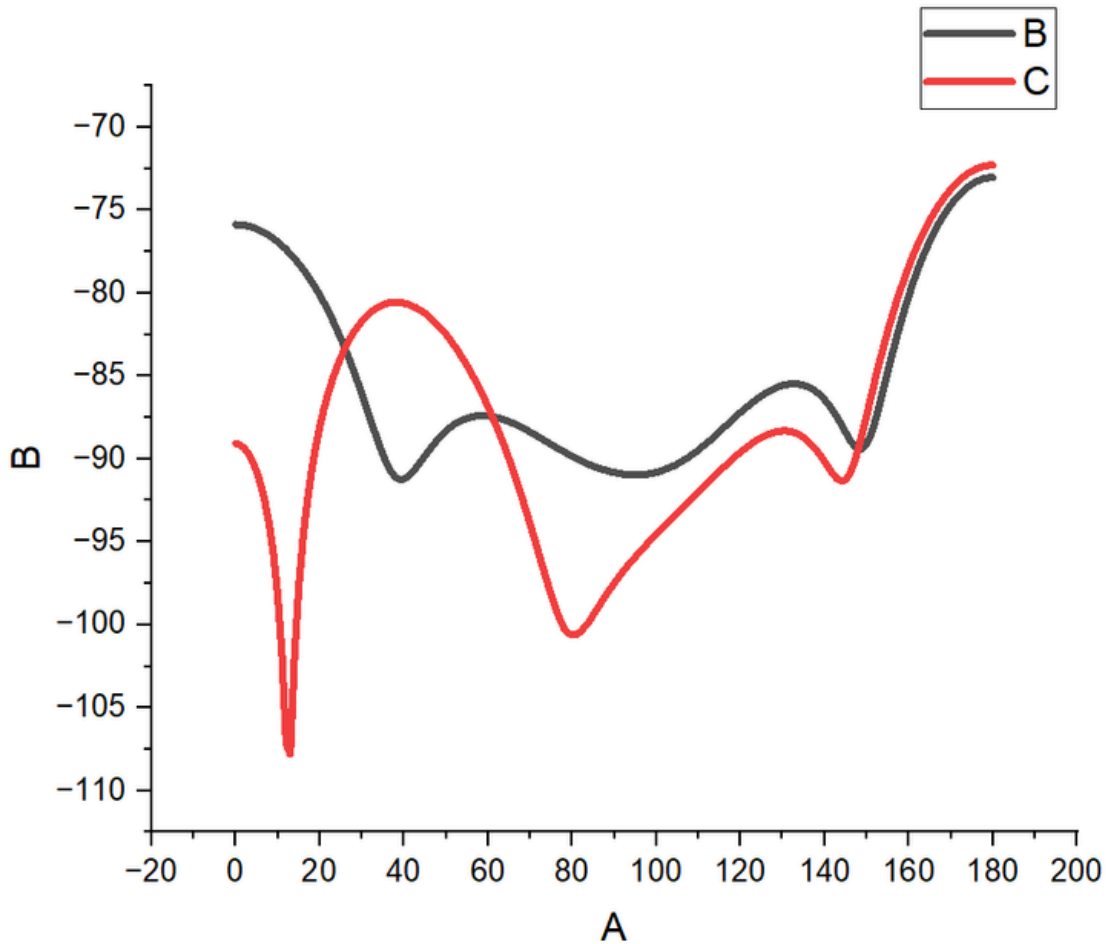
Fig.[10] This is shows the RCS of Metamaterial Unit cell

- This graph displays the bistatic radar cross section (RCS) of your graphene-based metamaterial array at a specific resonant frequency (15.7143 THz).
- Unlike the uniform metal reference (PEC), this curve maps out exactly how your engineered unit cells alter, absorb, and suppress scattered electromagnetic fields across different observation angles.

Model Review

- RCS(Radar Cress Section):-** To evaluate a metamaterial's performance, simulate a PEC plate of the identical physical dimensions as a baseline. The difference in their scattered fields, calculated as $RCS(pec)-RCS(meta)$, directly quantifies the metamaterial's RCS reduction.

Fig.[10] This is shows the RCS of Metamaterial Unit cell



[11]

Scattering Angle	PEC RCS	Metamaterial RCS	Net RCS Reduction	Performance status
0°	-76 dB(m^2)	-89 dB(m^2)	13 dB	Excellent Suppression
14°	-82 dB(m^2)	-107 dB(m^2)	25 dB	Maximum Peak Suppression
80°	-90 dB(m^2)	-100 dB(m^2)	10 dB	Broad Absorption Window
180°	-73 dB(m^2)	-72.3 dB(m^2)	None	Main Specular Lobe

CONCLUSION

Metasurface-based absorbers are a game-changer for 5th-generation aircraft, significantly enhancing stealth by reducing radar cross-section (RCS). These absorbers use engineered electromagnetic properties to efficiently absorb radar waves, minimizing reflections and improving survivability in modern warfare.

Recent research has focused on radar-absorbing grilles (RAGs) installed at aircraft engine inlets, which allow airflow while absorbing radar waves. By incorporating indium tin oxide (ITO) films, scientists have optimized impedance characteristics, achieving broadband absorption and high angular stability.

Overall, metasurface absorbers provide lightweight, adaptable, and highly effective stealth solutions, making them a promising technology for next-generation aircraft designs.

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2. Bei Bei. Xing , Zhen Guo. Liu , Wei Bing. Lu , and Hao. Chen. State Key Laboratory of Millimeter Waves, School of Information Science and Engineering, Southeast University, Nanjing, China Synergetic Innovation Center of Wireless Communication Technology, Southeast University, Nanjing, China
3. Ghosh, S. K., Das, S., & Bhattacharyya, S. (2019). A Graphene Based Broadband Metasurface Absorber in the Terahertz Region. 2019 URSI Asia-Pacific Radio Science Conference (AP-RASC).
4. Ghosh, S. K., Bhattacharyya, S., & Das, S. (2020). Graphene-based metasurface for wideband linear to circular polarization conversion. 2020 URSI Regional Conference on Radio.
5. Zhang, X., Xia, L., Zhang, Z., Zhang, X., Wei, D., Nie, C., ... Du, C. (2016). Terahertz amplitude modulator with graphene based metasurface. 2016 IEEE International Conference.
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THANK YOU!

